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LINUX FUNDAMENTAL ON HACK THE BOX

1.1 Introduction

I have been learning Linux Fundamentals to build a strong foundation in the Linux operating system, widely used in servers, cloud environments, and security operations. I have gained skills in basic shell commands, scripting, and system monitoring, which help me manage systems, enumerate services, and respond to security incidents effectively.

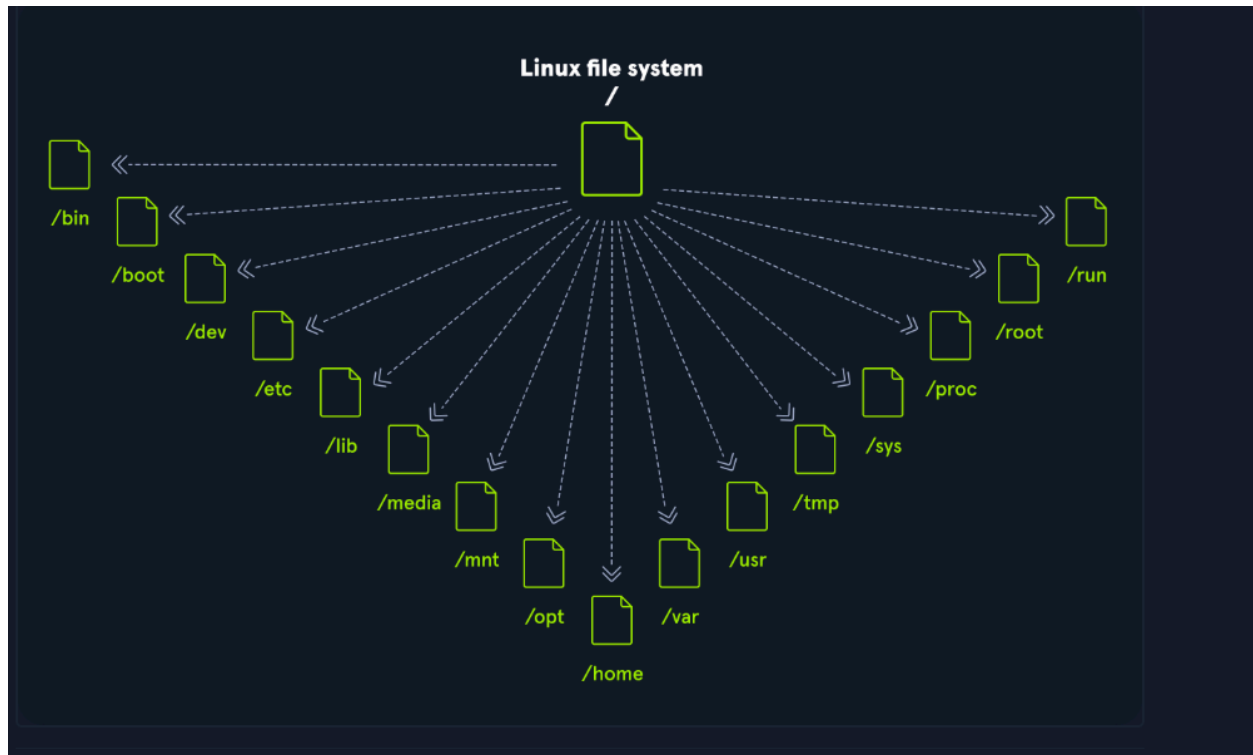
1.2 Linux Structure

Linux is an open source operating system that manages all the hardware resources of a computer. Linux comes in many distributions known as Distros. Some of the most popular and well-known being Ubuntu, Debian, Fedora, OpenSUSE, elementary, Manjaro, Gentoo Linux, RedHat, and Linux Mint. There are different components of Linux included in the diagram below:

Components	
Component	Description
Bootloader	A piece of code that runs to guide the booting process to start the operating system. Parrot Linux uses the GRUB Bootloader.
OS Kernel	The kernel is the main component of an operating system. It manages the resources for system's I/O devices at the hardware level.
Daemons	Background services are called "daemons" in Linux. Their purpose is to ensure that key functions such as scheduling, printing, and multimedia are working correctly. These small programs load after we booted or log into the computer.
OS Shell	The operating system shell or the command language interpreter (also known as the command line) is the interface between the OS and the user. This interface allows the user to tell the OS what to do. The most commonly used shells are Bash, Tcsh/Csh, Ksh, Zsh, and Fish.
Graphics server	This provides a graphical sub-system (server) called "X" or "X-server" that allows graphical programs to run locally or remotely on the X-windowing system.
Window Manager	Also known as a graphical user interface (GUI). There are many options, including GNOME, KDE, MATE, Unity, and Cinnamon. A desktop environment usually has several applications, including file and web browsers. These allow the user to access and manage the essential and frequently accessed features and services of an operating system.
Utilities	Applications or utilities are programs that perform particular functions for the user or another program.

1.2.1 Linux File system

Linux has a tree like file system as illustrated in the diagram below:



1.2.2 Linux Distribution

Linux distributions, or distros, are operating systems built on the Linux kernel, each tailored for specific purposes such as servers, desktops, embedded devices, or mobile computing.. Popular examples include **Ubuntu, Fedora, CentOS, Debian, and Red Hat Enterprise Linux**, with **Kali Linux** being widely used in cybersecurity for its security-focused tools.

1.3 Introduction to Shell

A Linux terminal, also called a shell or command line, provides a text-based input/output (I/O) interface between users and the kernel for a computer system. The most commonly used shell in Linux is the Bourne-Again Shell (BASH), and is part of the GNU project.

1.4 Prompt Description

They simply show information like your username, the computer name and the directories or folders you are currently using. Other special characters are as illustrated in the diagram below:

Special Character	Description
\d	Date (Mon Feb 6)
\D{%Y-%m-%d}	Date (YYYY-MM-DD)
\H	Full hostname
\j	Number of jobs managed by the shell
\n	Newline
\r	Carriage return
\s	Name of the shell
\t	Current time 24-hour (HH:MM:SS)
\T	Current time 12-hour (HH:MM:SS)
\@	Current time
\u	Current username
\w	Full path of the current working directory

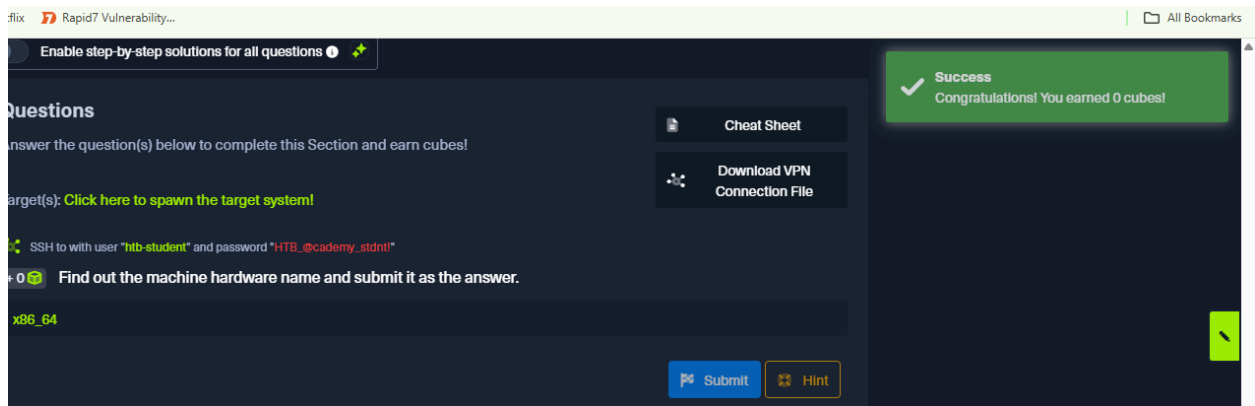
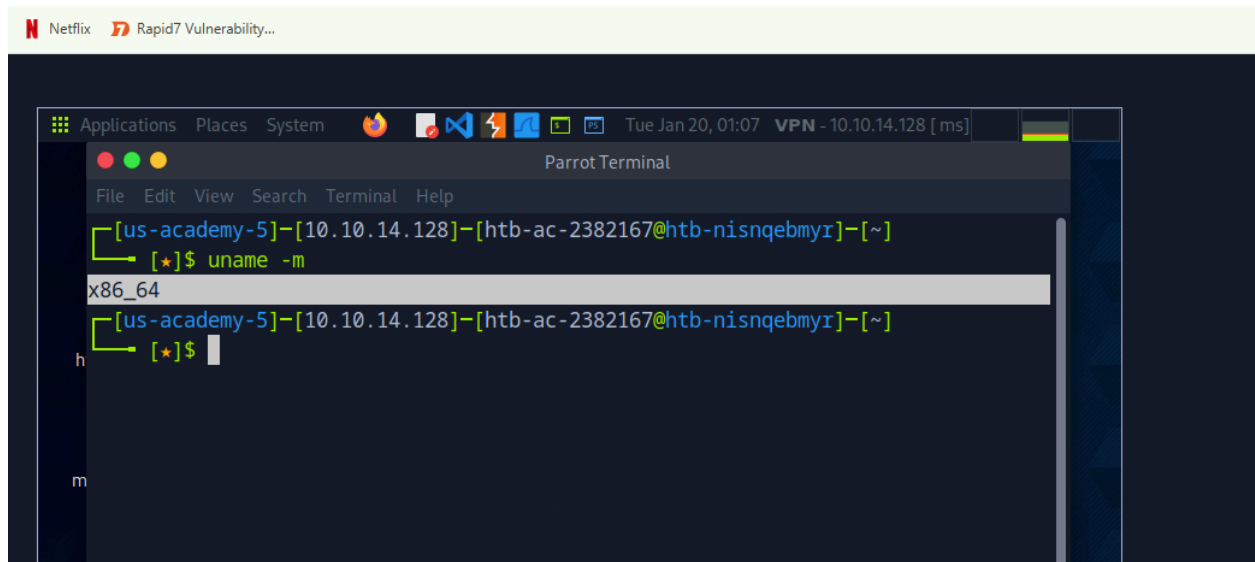
1.5 System Information

Question

1. Find out the machine hardware name and submit it as the answer.

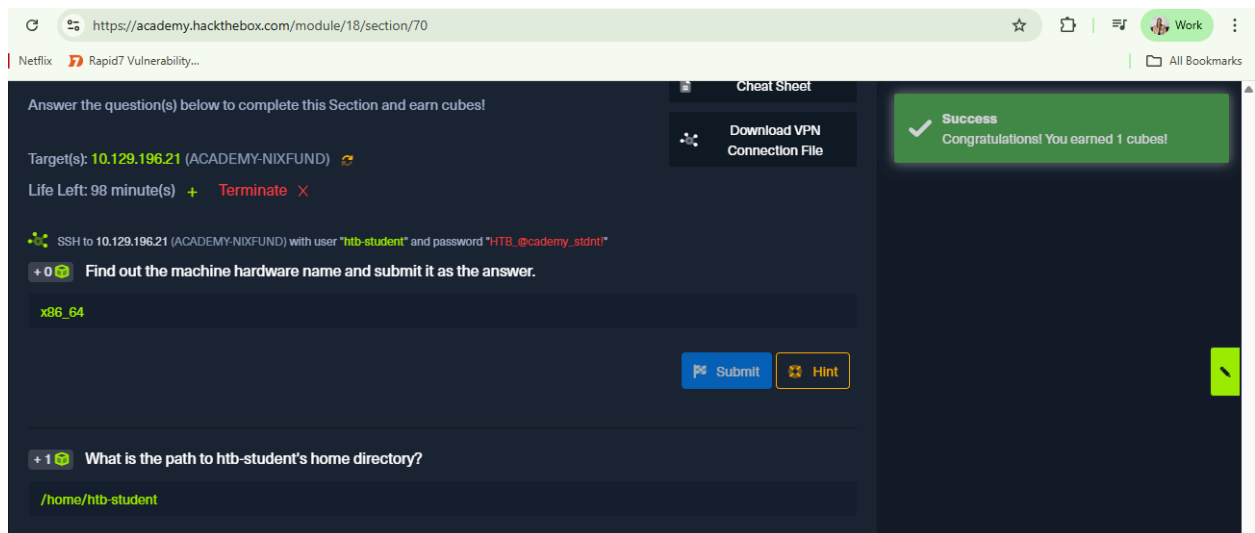
Answer: x86_64

Explanation : I got the machine hardware name by running the command “**uname -m**”



2. What is the path to htb-student's home directory?

Answer: /home/htb-student



```
kali@kali: ~/Downloads htb-student@nixfund: ~
Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 4.15.0-123-generic x86_64)
 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Tue Jan 20 08:29:54 UTC 2020

System load:  0.0          Processes:      148
Usage of /:   51.0% of 6.76GB Users logged in:  0
Memory usage: 20%         IP address for ens192: 10.129.11.254
Swap usage:   0%

 * Canonical Livepatch is available for installation.
   - Reduce system reboots and improve kernel security. Activate at:
     https://ubuntu.com/livepatch

0 packages can be updated.
0 updates are security updates.

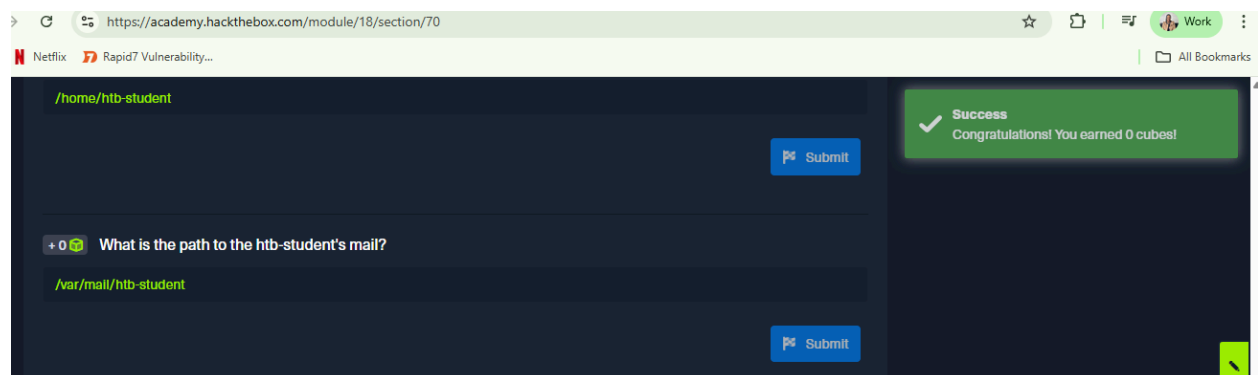
Last login: Wed Sep 23 22:09:41 2020 from 10.10.14.6
htb-student@nixfund:~$ /home/htb-student
-bash: /home/htb-student: Is a directory
```

3. What is the path to the htb-student's mail?

Answer: /var/mail/htb-student

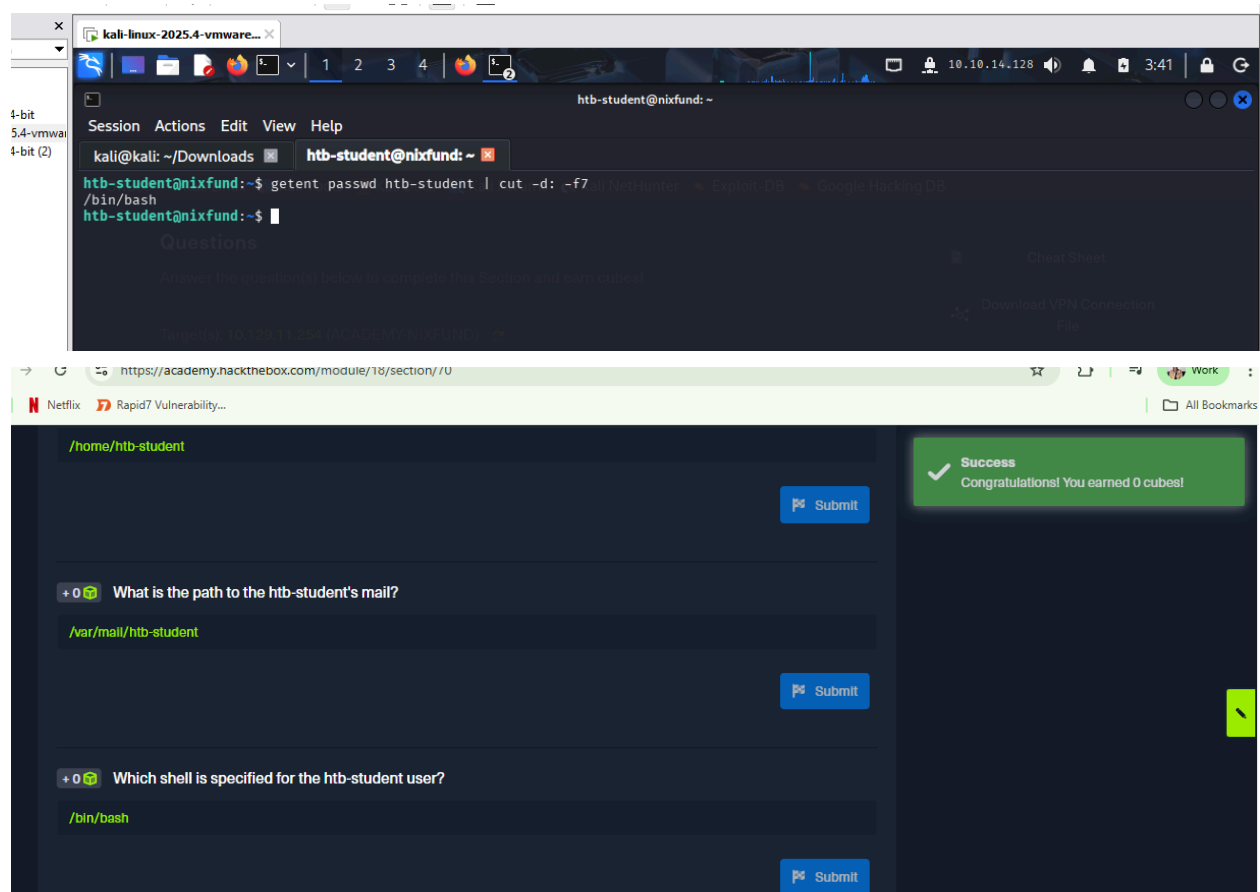
Explanation: After spawning the target, I ran the command echo \$MAIL.

```
htb-student@nixfund:~$ echo $MAIL
/var/mail/htb-student
htb-student@nixfund:~$
```



4. Which shell is specified for the htb-student user?

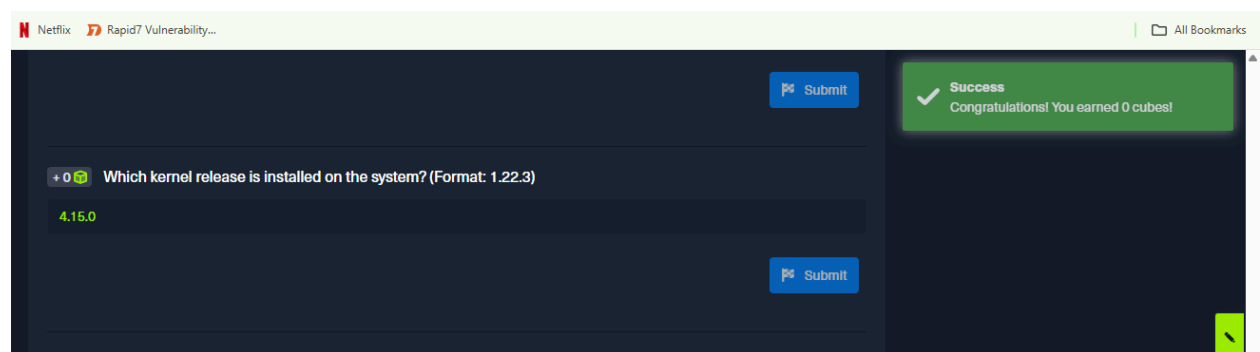
Answer: /bin/bash



5. Which kernel release is installed on the system? (Format: 1.22.3)

Answer:

4.15.0



```
htb-student@nixfund:~$ uname -r
4.15.0-123-generic
htb-student@nixfund:~$
```

6. What is the name of the network interface that MTU is set to 1500?

Answer: ens192

Explanation ; i used the command ifconfig.

```
htb-student@nixfund:~$ ifconfig
ens192: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.129.11.254 netmask 255.255.0.0 broadcast 10.129.255.255
    inet6 dead:beef::250:56ff:feb0:1caf prefixlen 64 scopeid 0<global>
    inet6 fe80::250:56ff:feb0:1caf prefixlen 64 scopeid 0<link>
    ether 00:50:56:b0:1c:af txqueuelen 1000 (Ethernet)
    RX packets 13479 bytes 992608 (992.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1126 bytes 95935 (95.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 2415 bytes 190234 (190.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2415 bytes 190234 (190.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

https://academy.hackthebox.com/module/18/section/70

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Submit

Success
Congratulations! You earned 1 cube!

0 Which kernel release is installed on the system? (Format: 1.22.3)
4.15.0
Submit

1 What is the name of the network interface that MTU is set to 1500?
ens192
Submit

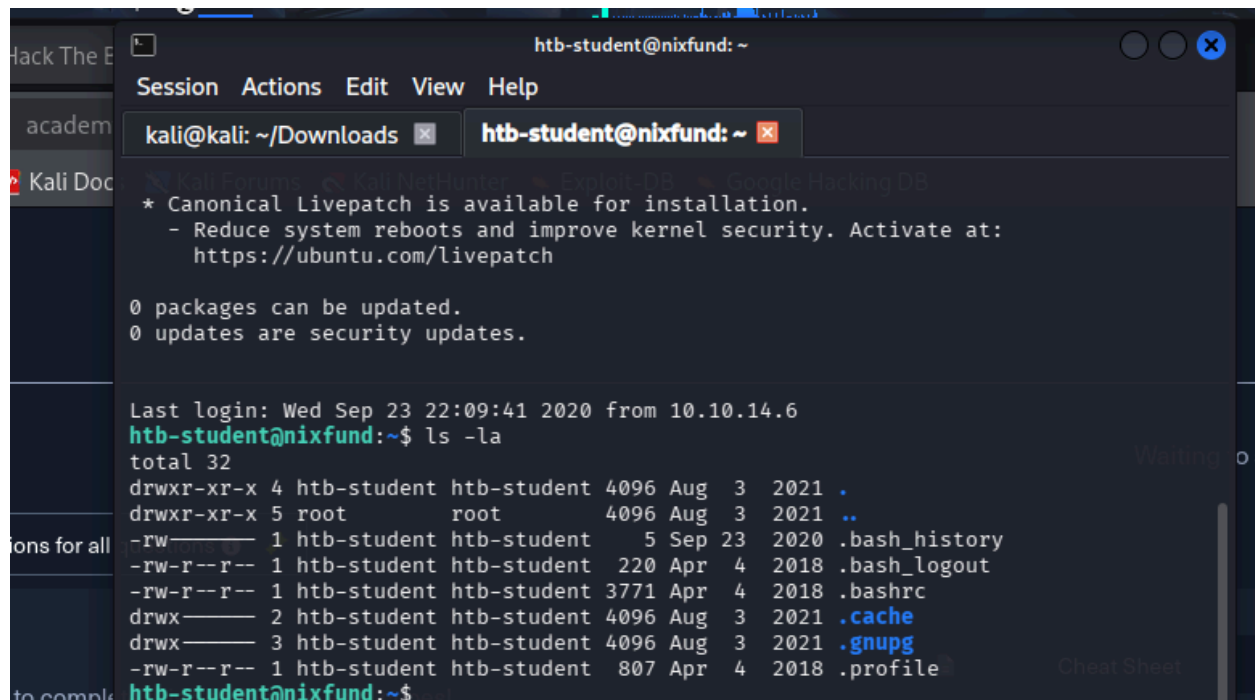
1.6 Navigation

Basically how to navigate the linux console. use `pwd`, to know in which directory you are in. `ls` list all the directories and `ls-la` list all the hidden files. `ls -l` list more information on those directories.

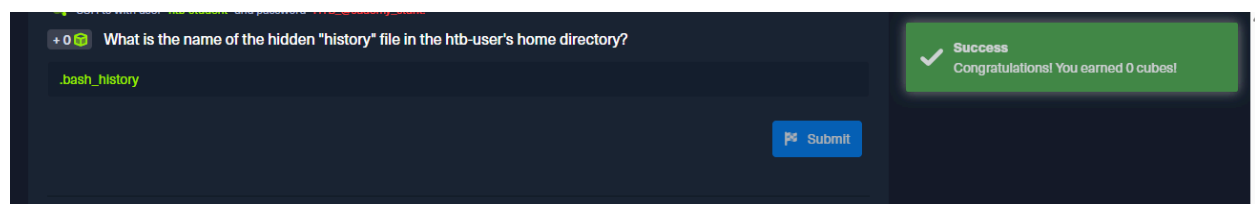
Question

What is the name of the hidden "history" file in the htb-user's home directory?

Answer: `bash_history`



```
htb-student@nixfund: ~  
Session Actions Edit View Help  
kali@kali: ~/Downloads x htb-student@nixfund: ~ x  
* Canonical Livepatch is available for installation.  
  - Reduce system reboots and improve kernel security. Activate at:  
    https://ubuntu.com/livepatch  
  
0 packages can be updated.  
0 updates are security updates.  
  
Last login: Wed Sep 23 22:09:41 2020 from 10.10.14.6  
htb-student@nixfund:~$ ls -la  
total 32  
drwxr-xr-x 4 htb-student htb-student 4096 Aug  3  2021 .  
drwxr-xr-x 5 root        root        4096 Aug  3  2021 ..  
-rw-r--r-- 1 htb-student htb-student  5 Sep 23  2020 .bash_history  
-rw-r--r-- 1 htb-student htb-student  220 Apr  4  2018 .bash_logout  
-rw-r--r-- 1 htb-student htb-student 3771 Apr  4  2018 .bashrc  
drwx----- 2 htb-student htb-student 4096 Aug  3  2021 .cache  
drwx----- 3 htb-student htb-student 4096 Aug  3  2021 .gnupg  
-rw-r--r-- 1 htb-student htb-student  807 Apr  4  2018 .profile  
htb-student@nixfund:~$
```



+0 🧐 What is the name of the hidden "history" file in the htb-user's home directory?

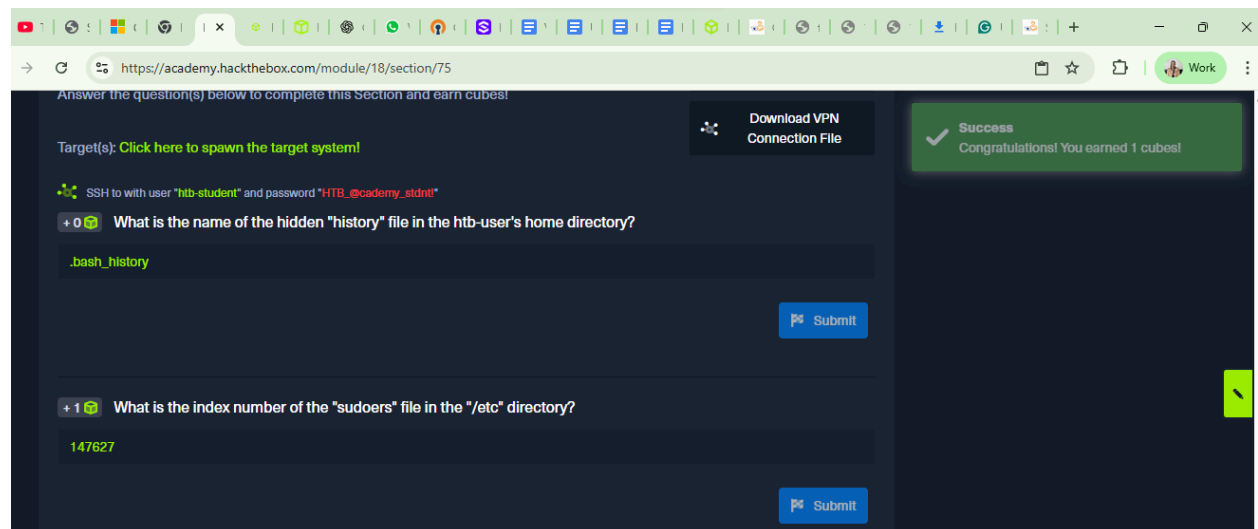
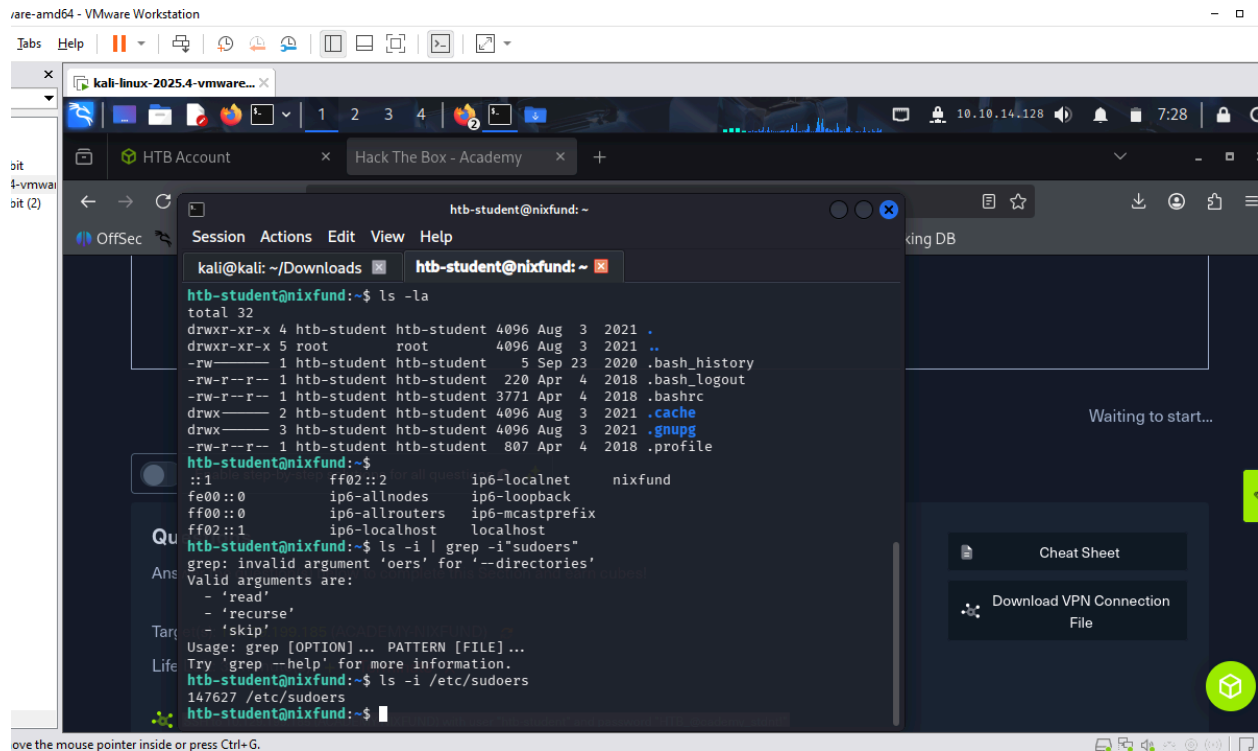
`.bash_history`

Submit

✓ Success
Congratulations! You earned 0 cubes!

2. What is the index number of the "sudoers" file in the "/etc" directory?

Answer: 147627



1.7 Working with files and Directories

In Linux, files and directories are managed mainly through the terminal, which allows creating (**touch**), organizing (**mkdir**), moving (**mv**), and copying (**cp**) files efficiently. Commands like **tree** help visualize structures, and features like redirection and automation make batch operations faster. Mastering these tools streamlines system management and boosts efficiency compared to using graphical interfaces.

Question

What is the name of the last modified file in the `"/var/backups"` directory?

Answer: `apt.extended_states.0`

The screenshot shows a Hack The Box Academy challenge interface. On the left, a terminal window displays the following commands and output:

```
htb-student@nixfund:~$ ls -t /var/backups | head -n 1
apt.extended_states.0
htb-student@nixfund:~$ ls -la -t
total 32
drwxr-xr-x 2 htb-student htb-student 4096 Aug 3 2021 .cache
drwxr-xr-x 3 htb-student htb-student 4096 Aug 3 2021 .gnupg
drwxr-xr-x 4 htb-student htb-student 4096 Aug 3 2021 ..
drwxr-xr-x 5 root root 4096 Aug 3 2021 .
-rw-r--r-- 1 htb-student htb-student 5 Sep 23 2020 .bash_history
```

On the right, a green success message box states: "Success Congratulations! You earned 0 cubes!". Below the terminal, a "Submit" button is visible.

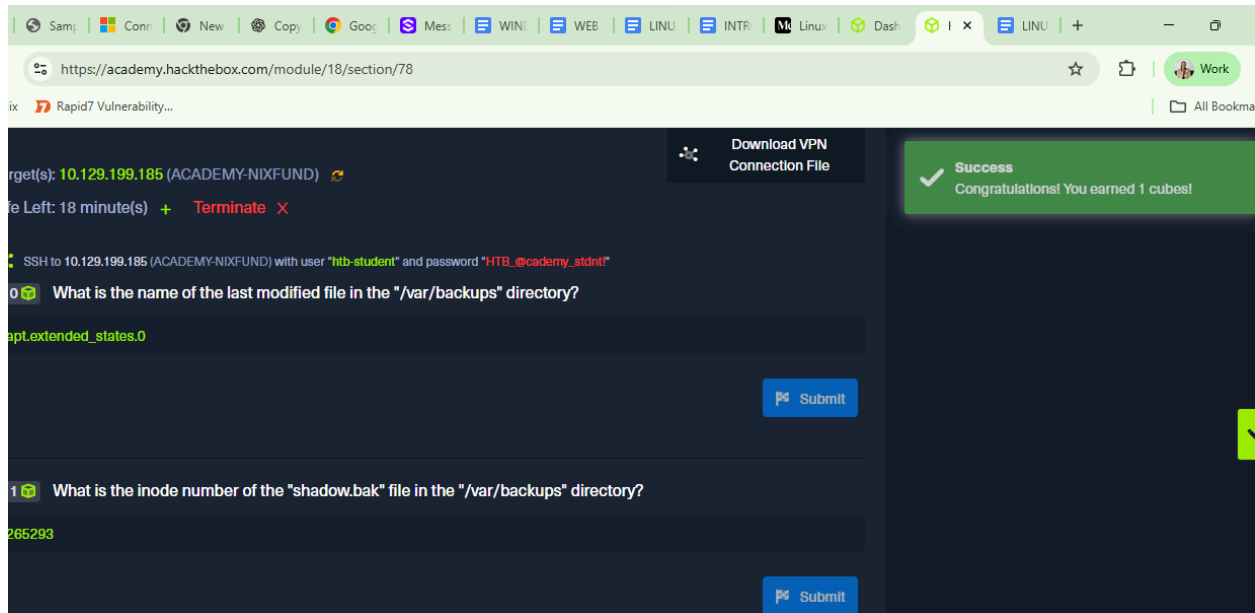
What is the inode number of the "shadow.bak" file in the "/var/backups" directory?

Answer: 265293

The screenshot shows a terminal window with the following commands and output:

```
htb-student@nixfund:~$ ls -li /var/backups/shadow.bak
265293 /var/backups/shadow.bak
```

The terminal output shows the inode number 265293 for the file /var/backups/shadow.bak.



1.8 Editing Files

Linux files can be edited using Nano for simplicity or Vim for advanced control, and managing key system files like `/etc/passwd` and `/etc/shadow` is essential for security.

1.9 Find Files and Directories

Finding files and folders in Linux is essential for locating configuration files, scripts, and other important data. The command shows the path of a program, e.g., `which python` returns `/usr/bin/python`. If the program doesn't exist, no result appears. Example: `find / -type f -name *.conf -user root -size +20k -newermt 2020-03-03 -exec ls -al {} \; 2>/dev/null`. For faster searches, `locate` uses a database updated with `sudo updatedb`, e.g., `locate *.conf`, but it offers fewer filters than `find`.

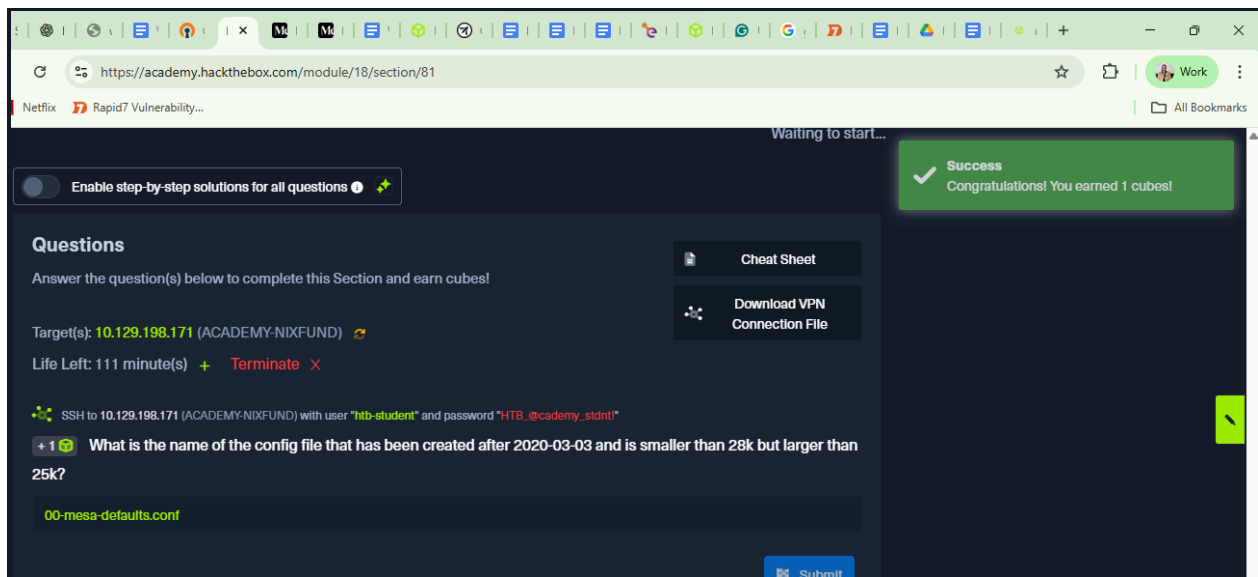
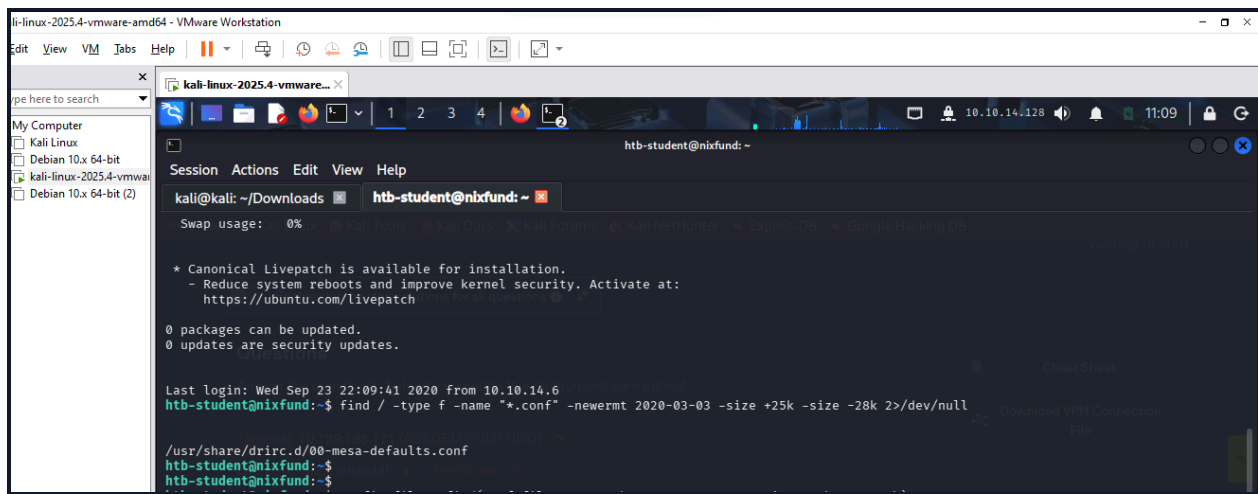
Question

What is the name of the config file that has been created after 2020-03-03 and is smaller than 28k but larger than 25k?

Answer:

00-mesa-defaults.conf

Explanation: I used this command to find the file including when it was created and what size. `find / -type f -name "*.conf" -newermt 2020-03-03 -size +25k -size -28k 2>/dev/null`

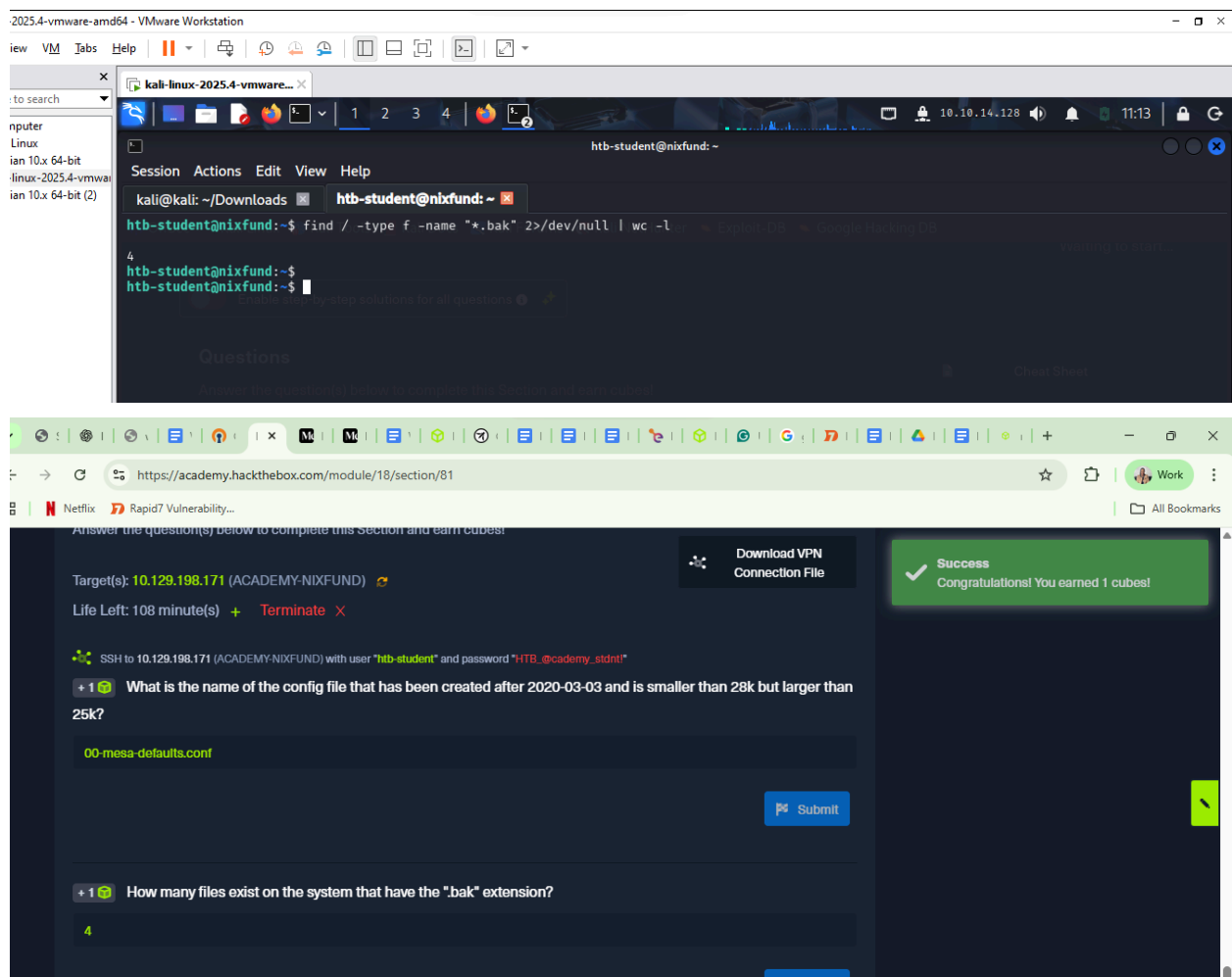


How many files exist on the system that have the ".bak" extension?

Answer:

4

Explanation: I ran this command on my target machine: `find / -type f -name "*.bak" 2>/dev/null | wc -l`



Submit the full path of the "xxd" binary.

Answer /usr/bin/xxd

The image shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the Ubuntu 18.04.5 LTS login screen for the user 'htb-student@nixfund'. The user has executed the command 'which xxd', which returned the output '/usr/bin/xxd'. Below the terminal, a web browser window shows the HackTheBox challenge page for '25k?'. The challenge is titled '00-mesa-defaults.conf' and asks the user to submit the full path of the 'xxd' binary. The user has entered '/usr/bin/xxd' and submitted it. A success message on the right side of the page states: 'Success Congratulations! You earned 0 cubes!'.

```
kali@kali: ~/Downloads htb-student@nixfund: ~ htb-student@nixfund: ~
Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 4.15.0-123-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Mon Jan 19 16:22:44 UTC 2026

System load:  0.02          Processes:           152
Usage of /:   51.0% of 6.76GB Users logged in:       1
Memory usage: 24%          IP address for ens192: 10.129.198.171
Swap usage:   0%

 * Canonical Livepatch is available for installation.
   - Reduce system reboots and improve kernel security. Activate at:
     https://ubuntu.com/livepatch

0 packages can be updated.
0 updates are security updates.

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Mon Jan 19 15:54:19 2026 from 10.10.14.128
htb-student@nixfund:~$
htb-student@nixfund:~$ which xxd
/usr/bin/xxd
htb-student@nixfund:~$
```

https://academy.hackthebox.com/module/18/section/81

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25k?

00-mesa-defaults.conf

Submit

+1 How many files exist on the system that have the ".bak" extension?

4

Submit

+0 Submit the full path of the "xxd" binary.

/usr/bin/xxd

Submit

Success
Congratulations! You earned 0 cubes!

2.0 File descriptors and redirections

A file descriptor is a system reference for managing input or output in Linux: 0 = STDIN (input), 1 = STDOUT (output), 2 = STDERR (errors). You can redirect errors with `2>/dev/null`, output to a file with `> filename` or append with `>> filename`, separate output and errors with `1>` and `2>`, read input from a file with `< filename`, or use a here-document with `<< EOF ... EOF`. Pipes (`|`) send the STDOUT of one command to another, e.g., `find /etc/ -name *.conf 2>/dev/null | grep systemd | wc -l`.

Questions

1 How many files exist on the system that have the ".log" file extension?

Answer :32

The image shows two screenshots. The top screenshot is a terminal window from a Kali Linux VM. It shows the command `find / -type f -name "*.log" 2>/dev/null | wc -l` being executed, which returns the output `32`. The bottom screenshot is a web browser showing the HackTheBox challenge page for "ACADEMY-NIXFUND". The page displays the question: "How many files exist on the system that have the '.log' file extension?". The user has entered the answer "32" and submitted it. A success message is shown: "Success Congratulations! You earned 1 cubes!".

2. How many total packages are installed on the target system?

Answer: 737

I ran the below command as shown in the screenshot.

The image shows a Kali Linux terminal window and a HackTheBox challenge page. The terminal window is titled "kali-linux-2025.4-vmware-amd64 - VMware Workstation" and shows the command `dpkg -l | grep '^ii' | wc -l` being executed, resulting in the output `737`. The terminal also shows the IP address `10.10.14.128` and the user `htb-student@nixfund`. The HackTheBox challenge page is titled "Questions" and shows the target IP address `10.129.198.171` (ACADEMY-NIXFUND). The challenge asks: "How many total packages are installed on the target system?" and the answer is `737`. A success message on the right says "Success Congratulations! You earned 0 cubes!".

kali-linux-2025.4-vmware-amd64 - VMware Workstation

Session Actions Edit View Help

htb-student@nixfund: ~

```
htb-student@nixfund:~$ dpkg -l | grep '^ii' | wc -l
737
htb-student@nixfund:~$
```

10.10.14.128

11:29

https://academy.hackthebox.com/module/18/section/79

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Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): 10.129.198.171 (ACADEMY-NIXFUND)

Life Left: 82 minute(s) + Terminate X

SSH to 10.129.198.171 (ACADEMY-NIXFUND) with user "htb-student" and password "HTB_@cademy_stdnt!"

+ 1 How many files exist on the system that have the ".log" file extension?

32

Submit

+ 0 How many total packages are installed on the target system?

737

Submit

Cheat Sheet

Download VPN Connection File

Success Congratulations! You earned 0 cubes!

2.1 Filter Contents

Linux tools like **more/less** view files, **head/tail** show start/end lines, **sort** orders output, **grep** filters patterns, **cut** and **tr** extract/replace fields, **awk** selects columns, **sed** substitutes text, and **wc** counts lines. Combined in pipelines, they let you efficiently filter and process **/etc/passwd** data.

Questions

1. How many services are listening on the target system on all interfaces? (Not on localhost and IPv4 only)

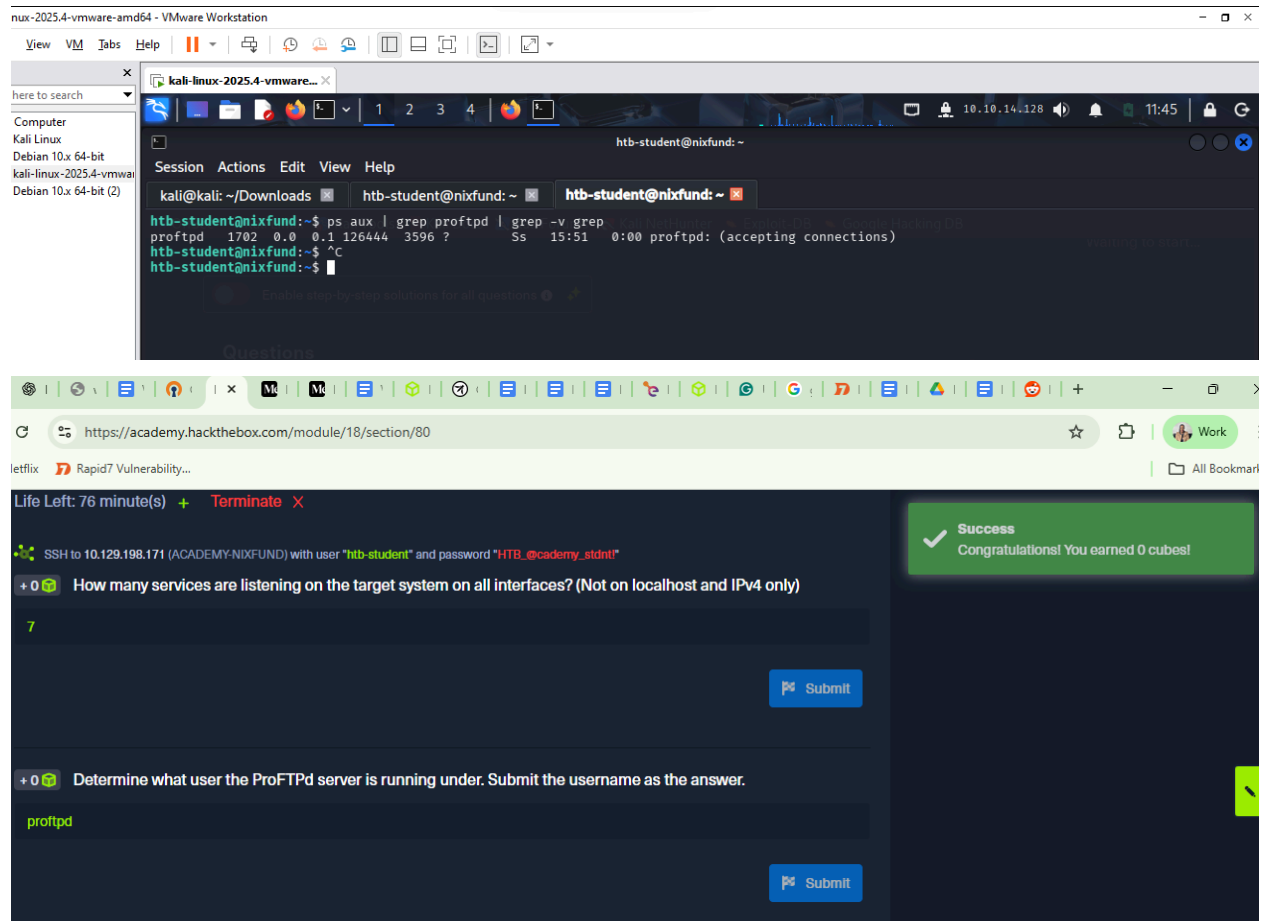
Answer : 7

The image shows a Kali Linux terminal window and a HackTheBox challenge interface. The terminal window displays the command `netstat -tlnp | grep -v tcp6 | grep -v "127.0.0.1" | grep -c LISTEN` and its output, which is 7. The HackTheBox interface shows the challenge details for "10.129.198.171 (ACADEMY-NIXFUND)" and the question: "How many services are listening on the target system on all interfaces? (Not on localhost and IPv4 only)". The answer "7" is entered in the input field, and a "Submit" button is visible. A success message on the right says "Success Congratulations! You earned 0 cubes!".

2. Determine what user the ProFTPD server is running under. Submit the username as the answer.

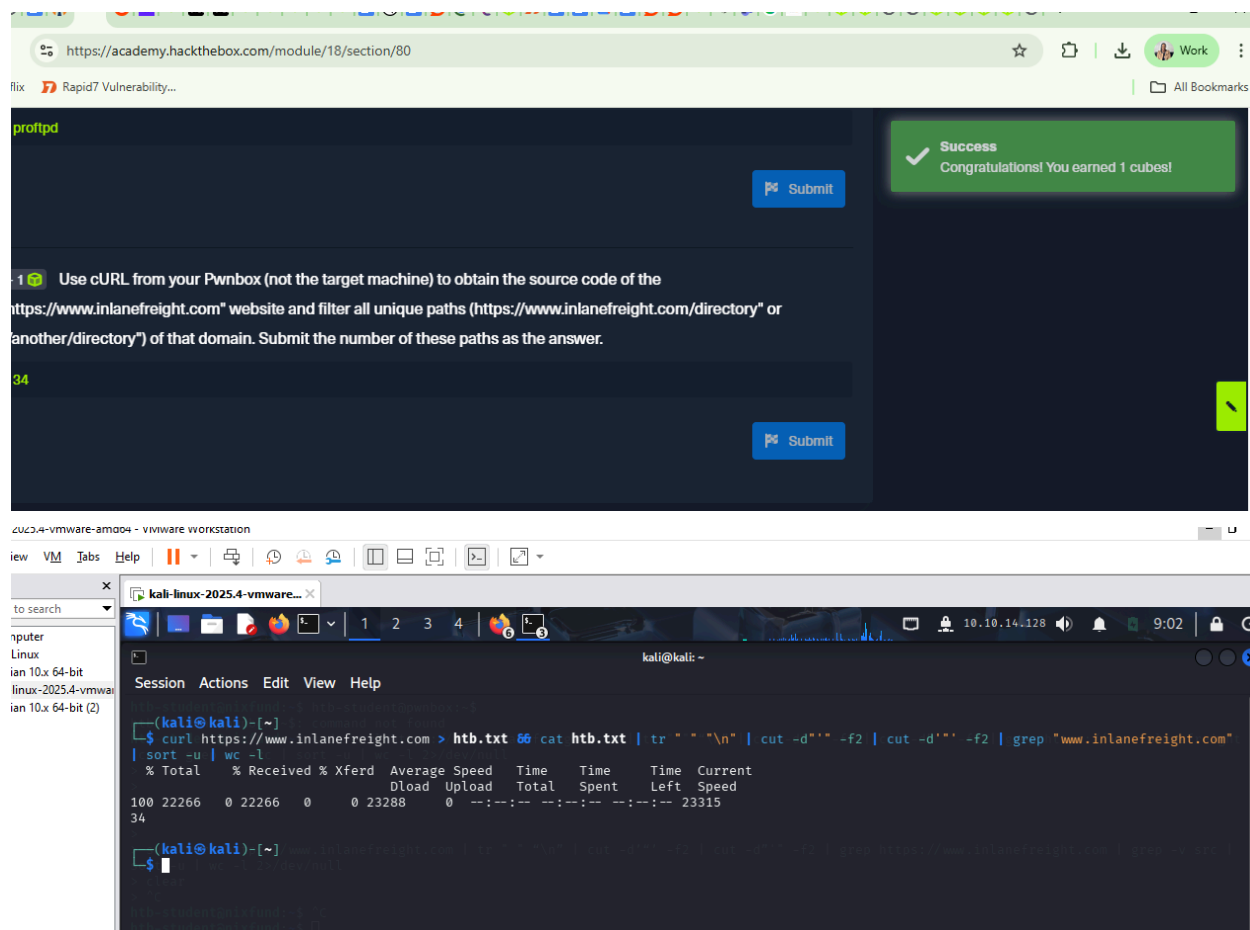
Answer

proftpd



3. Use cURL from your Pwnbox (not the target machine) to obtain the source code of the "https://www.inlanefreight.com" website and filter all unique paths (https://www.inlanefreight.com/directory" or "/another/directory") of that domain. Submit the number of these paths as the answer.

Answer: 34



2.2 Regular Expressions

Regular expressions (RegEx) are patterns for searching and manipulating text using symbols like `()` for grouping, `[]` for character classes, `{ }` for repetition, and `|` for OR. Tools like `grep` use RegEx to filter text, e.g., finding lines with one or multiple patterns, ignoring comments, or matching specific settings in files like `sshd_config`.

2.3 Permission Management

Linux permissions control who can read, write, or execute files and directories. `chmod` changes permissions, `chown` changes ownership, octal values represent permission sets, SUID/SGID allow running programs with owner/group rights, and the sticky bit restricts file deletion in shared directories.

2.4 User Management

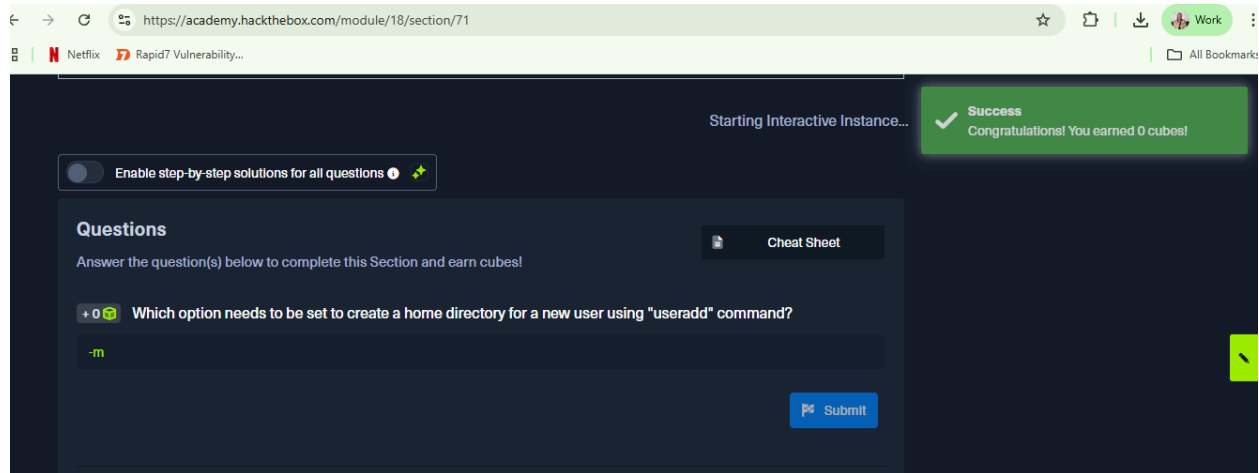
Linux user management lets you create users, assign groups, and run commands with proper privileges using `sudo` or `su`. Commands like `useradd`, `usermod`, `userdel`, and `passwd` control access, helping maintain system security.

Question

1. Which option needs to be set to create a home directory for a new user using "useradd" command?

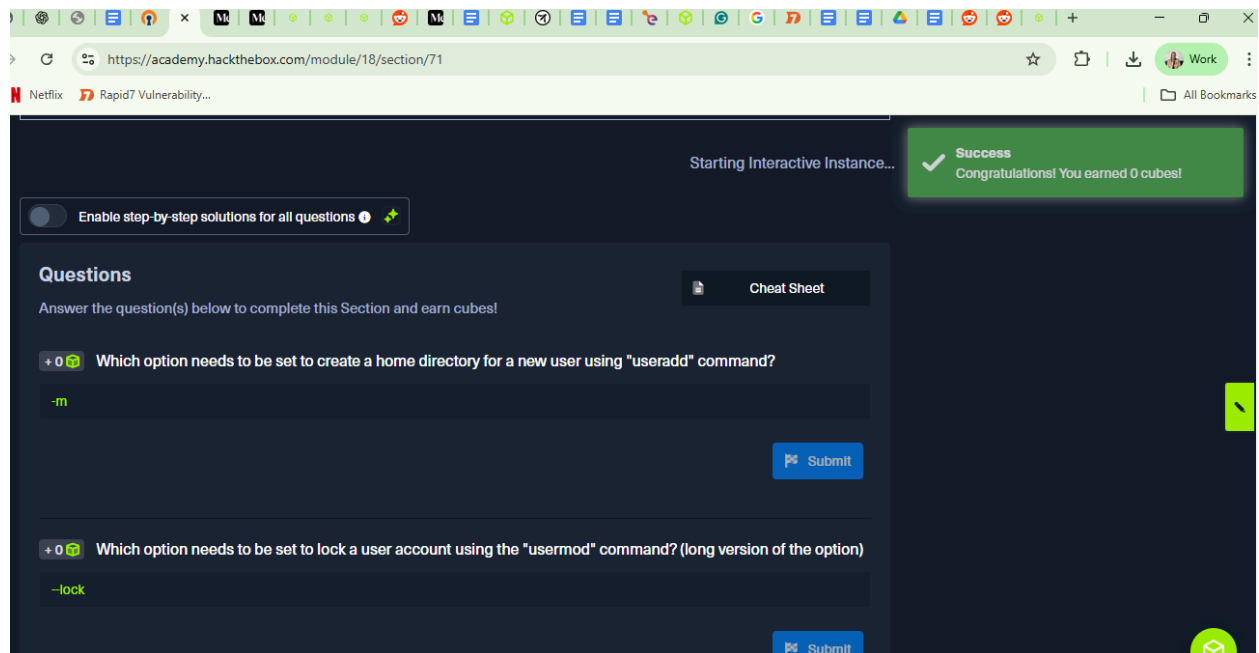
Answer: `-m`

Short for "make home"



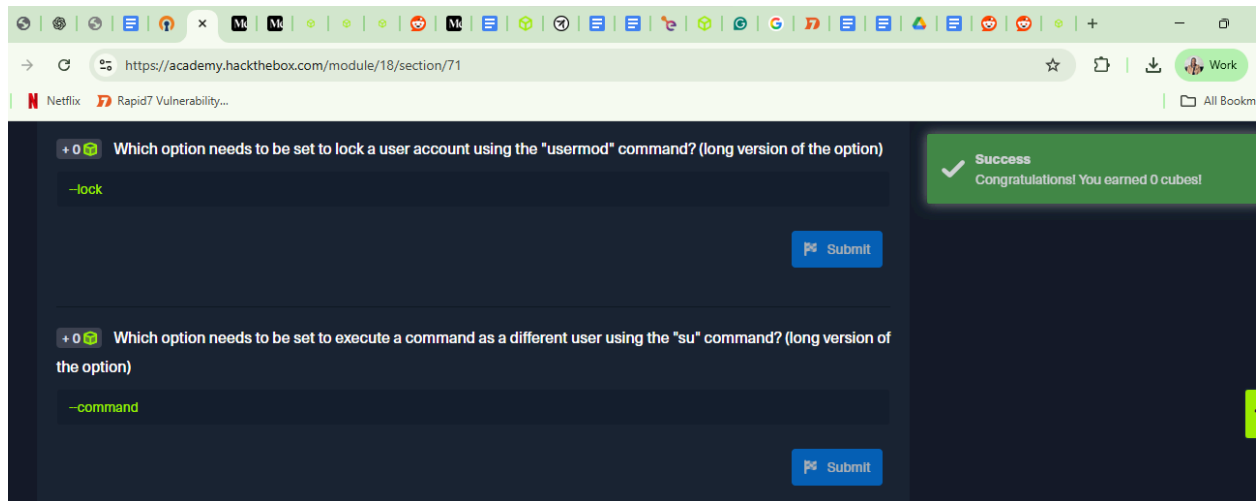
2. Which option needs to be set to lock a user account using the "usermod" command? (long version of the option)

Answer: `-lock`



3. Which option needs to be set to execute a command as a different user using the "su" command? (long version of the option)

Answer:--command



2.5 Package Management

Linux package managers let you install, update, and remove software while handling dependencies. Debian-based systems use **APT/dpkg**, Red Hat uses **yum/rpm**, and others include **snap**, **pip**, **gem**, and **git**. Commands like **apt install**, **apt-cache search**, and **dpkg -i** manage packages, while Git can clone projects from repositories.

2.6 Service and process management

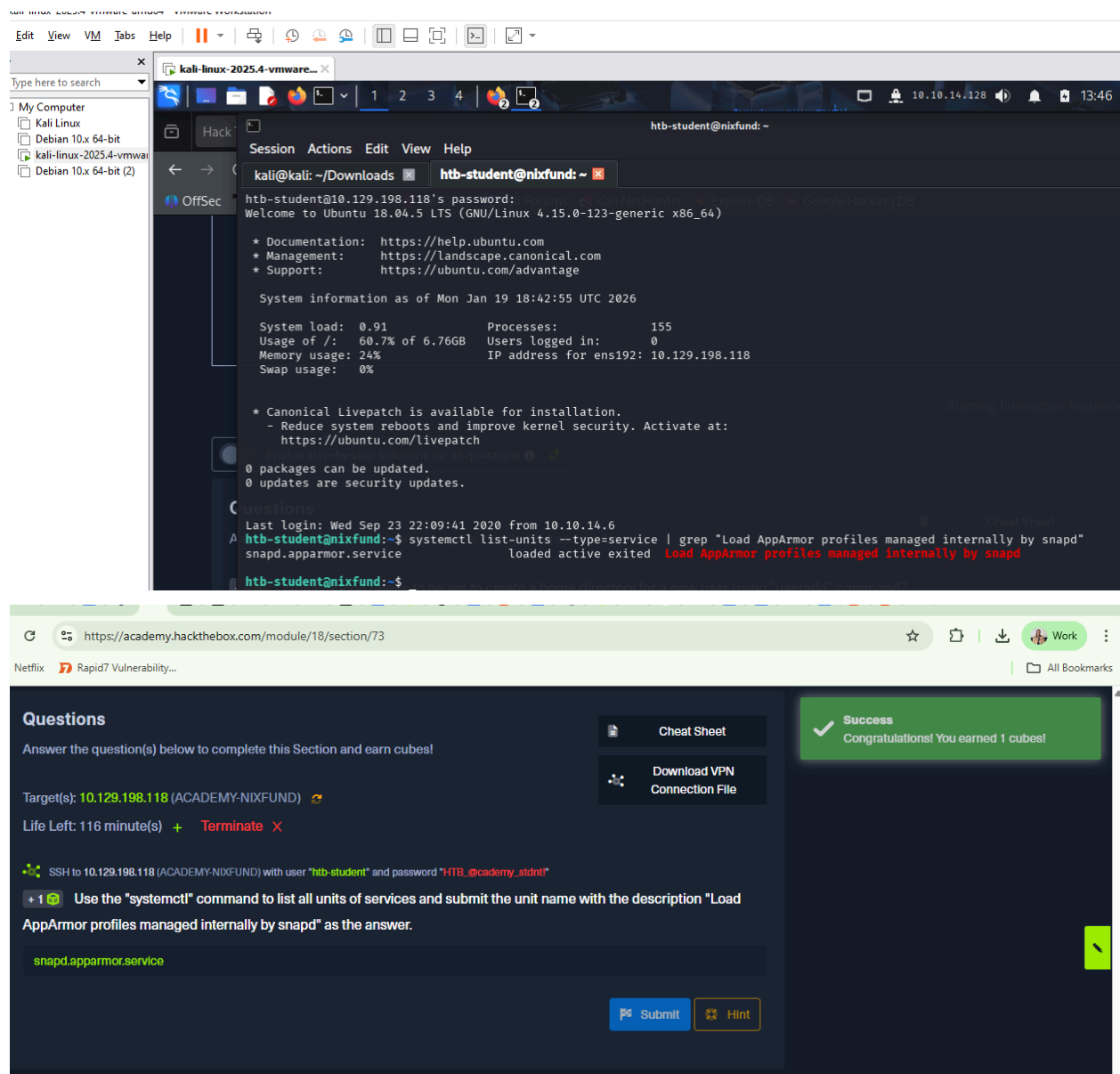
Linux services (daemons) run in the background to keep the system working, are managed with **systemctl** (start/stop/status/enable), monitored via **journalctl** and **ps**, controlled with signals using **kill**, and processes can be backgrounded or chained together using **&**, **;**, **&&**, and **|**.

Question

Use the "systemctl" command to list all units of services and submit the unit name with the description "Load AppArmor profiles managed internally by snapd" as the answer.

Answer

Snapd.apparmor.service



2.7 Task Scheduling

Linux automates tasks using **Systemd** (timers + services) or **Cron** (crontab). Systemd runs scripts at events or intervals; Cron runs scripts at set times. Both handle updates, backups, or maintenance automatically.

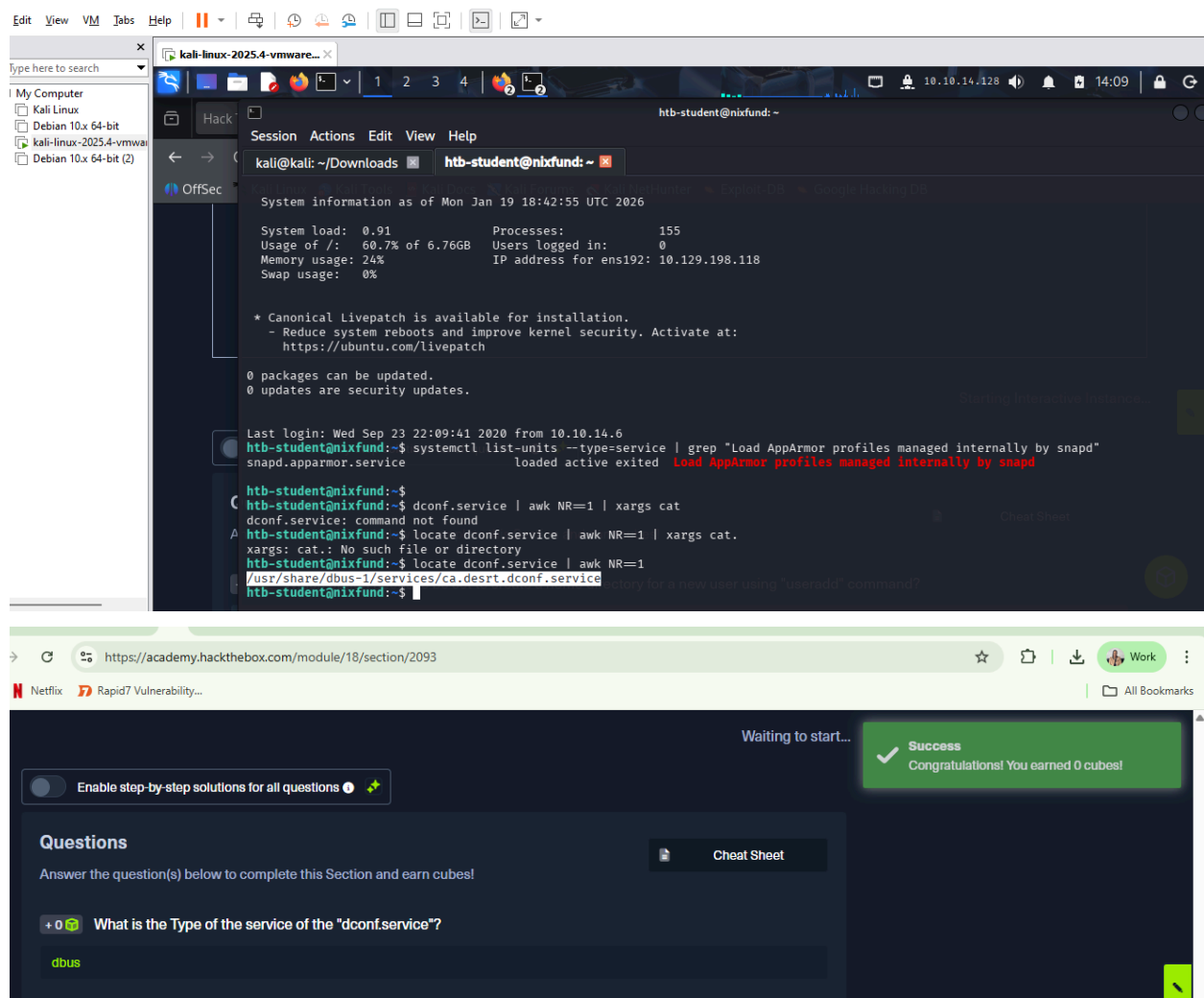
Question

What is the Type of the service of the "dconf.service"?

Answer

dbus

explanation: This service is defined as a **D-Bus activated service**, meaning it starts on demand via D-Bus rather than running continuously



2.8 Network Services

SSH, NFS, Web Server, and VPN are Linux network services used for secure remote access, file sharing, web hosting, and encrypted network connections; install/configure them via **apt** or Python (**openssh-server**, **nfs-kernel-server**, **apache2**, **python3 -m http.server**, **openvpn**) and manage settings in their respective config files for administration and pentesting.

2.9 Working with Web Servers

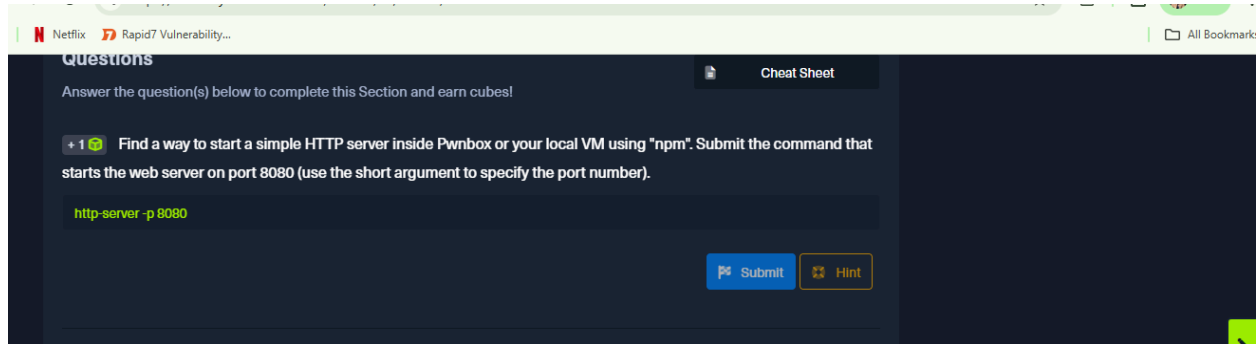
Web services let browsers and servers communicate; Apache hosts sites, supports modules (e.g., **mod_ssl**, **mod_proxy**) and dynamic content, runs on port 80 by default, and can be tested with **curl**, **wget**, or **python3 -m http.server**.

Questions

Find a way to start a simple HTTP server inside Pwnbox or your local VM using "npm". Submit the command that starts the web server on port 8080 (use the short argument to specify the port number).

Answer: `http-server -p 8080`

Explanation: `http-server -p 8080` starts a simple Node.js HTTP server and uses the **short -p option** to listen on **port 8080**, serving files from the **current directory**.



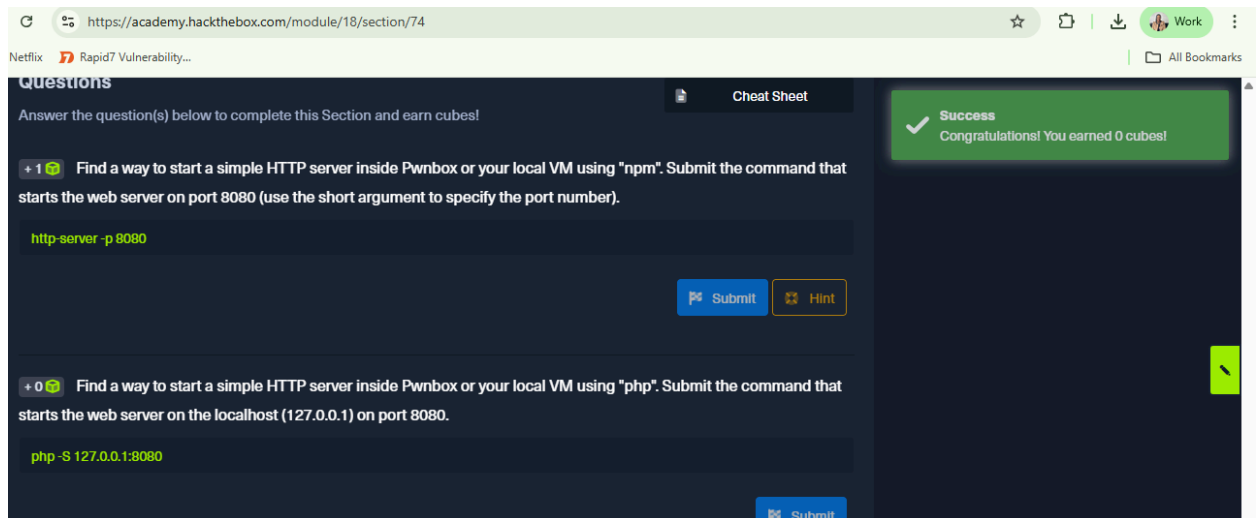
Question

Find a way to start a simple HTTP server inside Pwnbox or your local VM using "php". Submit the command that starts the web server on the localhost (127.0.0.1) on port 8080.

Answer: `php -S 127.0.0.1:8080`

Explanation:

This uses PHP's built-in development server (**-S**) to start a simple HTTP server bound to **localhost (127.0.0.1)** on **port 8080**, serving files from the current directory



3.0 Back up and Restore

Linux backup and restore can be done efficiently using tools like **rsync** (fast), **Duplicity** (encrypted), or **Deja Dup** (GUI). To back up a directory, use **rsync -avz -e ssh /source/**

`user@backup:/backup/` and to restore, `rsync -av user@backup:/backup/ /source/`. Backups can be automated with cron jobs.

3.1 File System Management

Linux file system management involves choosing appropriate file systems, organizing data using a hierarchical structure with inodes, managing disks and partitions, mounting/unmounting storage, and handling swap space to ensure performance, reliability, and efficient storage use.

Question

How many partitions exist in our Pwnbox? (Format: 0) :

Answer: 3

Explanation: I spawned and got the 3 partitions, vda1,vda2,vda5.

The screenshot displays a Parrot Terminal window with the following output from the `lsblk` command:

```
[us-academy-5]-[10.10.14.128]-[htb-ac-2382167@htb-nisnqebmyr]-[~]
[*]$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
vda         254:0    0   40G  0 disk
├─vda1      254:1    0   39G  0 part /
├─vda2      254:2    0    1K  0 part
└─vda5      254:5    0  975M  0 part [SWAP]
```

Below the terminal, a web browser window shows the HackTheBox Academy interface. A green success message states: "Success Congratulations! You earned 0 cubes!". The question section asks: "How many partitions exist in our Pwnbox? (Format: 0)". The answer "3" has been entered in the input field, and a "Submit" button is visible at the bottom right.

3.2 Containerization

Containerization packages applications with their dependencies into isolated, lightweight environments like Docker or LXC, enabling consistent, portable, and scalable deployment while requiring proper security measures.

3.3 Network Configuration.

As a penetration tester, mastering Linux network configuration, access control (SELinux, AppArmor, TCP wrappers), and monitoring tools is essential for securing systems, troubleshooting issues, and performing effective testing.

3.4 Remote desktop protocols in Linux

VNC and X11 let Linux admins access remote desktops; VNC is encrypted, X11 needs SSH tunneling for security. Administrators configure servers, set passwords, and manage display ports, ensuring safe and efficient remote management of Linux systems.

3.5 Linux Security

Linux security relies on updates, firewalls, least-privilege access, and tools like SELinux, AppArmor, and fail2ban. Unnecessary services should be removed, strong passwords enforced, and TCP wrappers used to control service access. Security is an ongoing process requiring careful configuration and monitoring.

3.6 Firewall Set up

Linux firewalls (iptables, nftables, UFW) control network traffic using tables, chains, rules, matches, and targets to allow, block, or log packets. They can filter by IP, port, or protocol, create custom chains, and manage traffic, providing flexible security for Linux systems.

3.7 System Logs

Linux system logs track activity for monitoring, troubleshooting, and security. Key logs include: **kernel** (/var/log/kern.log), **system** (/var/log/syslog), **auth** (/var/log/auth.log), **application** (e.g., Apache/MySQL), and **security** (Fail2ban/UFW). Regular review helps detect attacks, misconfigurations, and suspicious activity.

3.8 Solaris

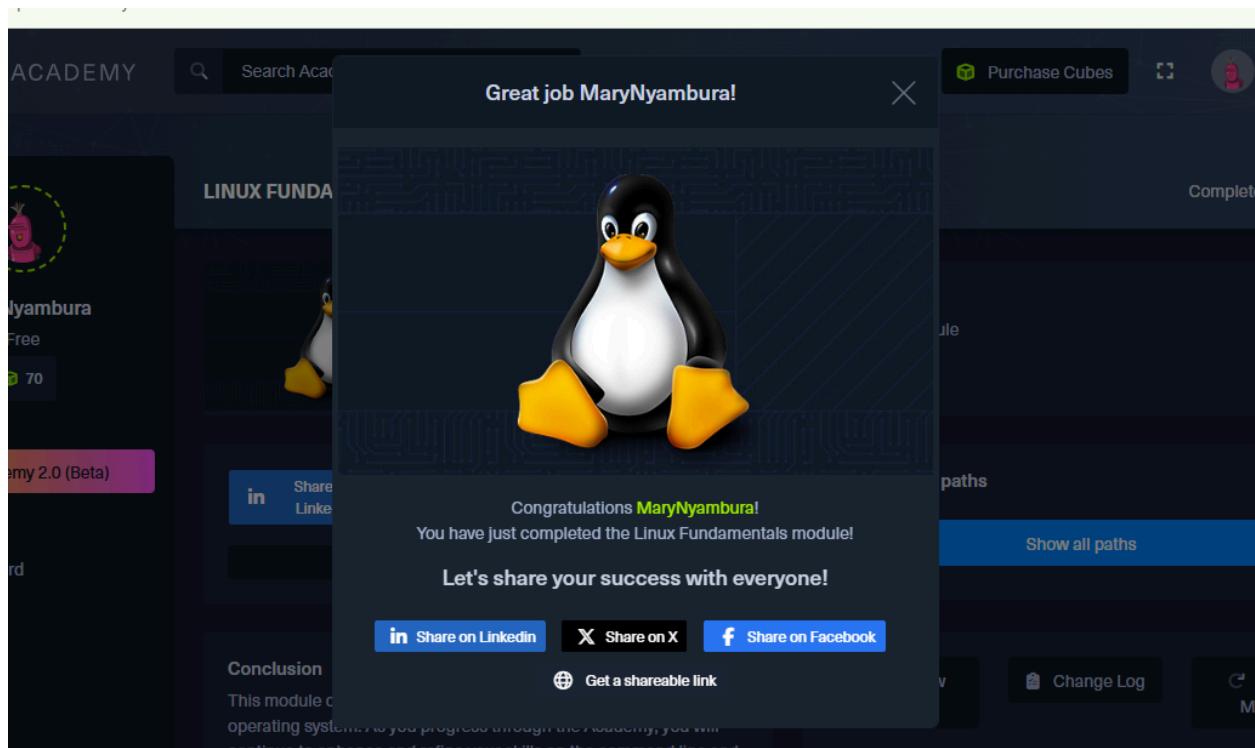
Solaris is a stable, scalable Unix-based OS used in enterprise environments. It features RBAC, SMF, and IPS package management, with proprietary commands for system info, packages, and

process tracing. Optimized for high-end hardware, virtualization, and secure, mission-critical applications.

Link for my completion of the module

<https://academy.hackthebox.com/achievement/2382167/18>

Screenshot displaying my completion of the module.



Conclusion

In learning Linux fundamentals, I've realized how important it is to understand how the system works from the ground up. From managing file systems and swap space, handling users and permissions, to backing up data securely with tools like `rsync` or automating tasks with `cron`, everything connects to keeping a system reliable and safe. I also see the power of containers with Docker and LXC, letting me create isolated environments to test or deploy applications without messing up my main system. Overall, knowing these basics makes me more confident in managing Linux, whether it's for personal projects or real-world environments.

